

Saatwik Tiwari

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EDUCATION

Birla Institute of Technology and Sciences, Goa Campus

B.E. (Hons.) Mathematics and Computing | **CGPA: 7.64**

Expected May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Linear Algebra, Discrete Mathematics, Probability and Statistics, Optimization, Object Oriented Programming, Database Management Systems

EXPERIENCE

Caarya Innovative

May 2026 – July 2026

Software Engineering Intern | *Production Systems & LLM Infrastructure*

- Architected and deployed a production-grade Root Cause Analysis (RCA) agent as a monolith backend serving 50+ enterprise beta users; achieved sub-1s response latency (p99 < 850ms) and sustained throughput of 200+ req/s through concurrent request handling, response streaming, and load-balanced LLM inference
- Engineered end-to-end system design integrating React frontend, Node.js/Express API layer with request queuing, FastAPI LLM backend with streaming inference, and PostgreSQL state persistence; implemented health checks, graceful degradation, and circuit-breaker patterns to ensure 99.2% uptime across production deployments
- Designed configurable LLM workflow engine supporting adaptive system-prompt templating, multi-model routing (Gemini, GPT-4, etc), and structured JSON extraction via prompt engineering; reduced hallucination rate by 40% through in-context learning and output validation
- Optimized vector search via Redis caching, reducing RCA generation time by 35% and DB pressure by 55%(17->8 queries)
- Built a comprehensive CI/CD pipeline with unit, integration, and E2E testing achieving 78% code coverage; automated Docker builds and GitHub Actions deployment with automated rollback on test failure [Live Demo](#)

TCS Research

Jan 2026 – Apr 2026

Student Researcher | *BITS Goa DataLab – LLM Alignment & Safety*

- Developed a quantitative sycophancy measurement framework evaluated across 8 open-source LLMs (Llama 2 70B, Mistral 7B, etc.) and 12 diverse domains (politics, science, ethics); designed multi-dimensional metrics (agreement score, confidence calibration, preference reversal rate) achieving 0.89 inter-annotator agreement on curated 500+ test samples
- Engineered a sycophancy mitigation pipeline combining automated prompt steering (contrastive prefix injection), LoRA fine-tuning (rank 16, $\alpha = 32$), and dynamic in-context learning; achieved 25% relative improvement over state-of-the-art baselines on sycophancy metrics while maintaining $\leq 2\%$ MMLU performance degradation
- Conducted rigorous ablation studies across 2 fine-tuning schedules, 6 prompt variations, and 2 inference strategies; validated improvements on held-out test set ($N = 200$) using statistical significance testing ($p < 0.05$, t-testing) and sensitivity analysis
- Authored research manuscript documenting evaluation protocols, experimental methodology, reproducibility details, and performance comparisons; prepared for submission to peer-reviewed venue; generated comprehensive ablation tables, learning curves, and error analysis supporting claims

PROJECTS

Crystal – AI Coding Assistant | *AI Systems, Software Engineering*

- Built an **AI coding harness** for real-time coding suggestions and repo-wide code generation/debugging via programming aligned LLMs; **indexed 1,000+ files/min** and achieved **sub-2s TTFT** using dependency graphs, semantic search, and RAG
- Debugged and optimized production bottlenecks in **FastAPI/Node.js** backends, attaining **sub-1s message latency**; implemented LLM-driven **Plan Mode** and testing blocks for reliable code generation and execution
- Engineered a FastAPI/Node.js backend combining REST APIs and WebSocket communication, supporting 10+ simultaneous users with low message latency [Live Demo](#) | [GitHub](#)

AIMvL – GitHub for ML Models | *Machine Learning Systems, MLOps*

- Implemented a full-stack AI model registry using React and Node.js, supporting PyTorch, TensorFlow, and ONNX models with versioning, training, inference, lifecycle management, and deployment through a unified interface
- Designed 15+ training recipes across 8+ architectures and engineered an asynchronous scheduler for concurrent GPU-accelerated jobs with Bayesian hyperparameter optimization, achieving 10–30% performance gains over baseline configurations
- Built an end-to-end reproducible ML pipeline with dataset versioning, artifact storage, and evaluation dashboards, reducing model deployment time from 1 hour to 5 minutes [Live Demo](#) | [GitHub](#)

OptiMover – Multiplayer Game Platform & Adaptive AlphaZero | *Full-stack, RL, Game AI*

- Architected a full-stack **multiplayer board-game platform** with **Prisma ORM**, persistent sessions, Elo rankings, match history and multi-level AI agents; tested with **20+ concurrent users** achieving **sub-100ms real-time update latency**
- Integrated an AlphaZero-based reinforcement learning system using Monte Carlo Tree Search with policy/value networks for strategic board games, achieving 52–74% win rates against human players and baseline minimax algorithms
- Addressed cold-start inefficiency through warm-start initialization with smaller CNN policy networks, reducing sample complexity and accelerating MCTS convergence versus standard self-play from scratch
- Engineered state-dependent MCTS depth allocation, dynamically adapting rollout budgets and network capacity across domains such as Connect-4 and Chess for improved compute efficiency and convergence
- Implemented an Elo-based ranking system with persistent users, matches, ratings, and game history while achieving sub-100ms real-time update latency [Live Demo](#) | [GitHub](#)

Adaptive Multi-Horizon Decision Transformer (AMH-DT) | *Reinforcement Learning, Finance*

- Developed hierarchical **RL** for multi-asset forecasting with **hypergraph-based dependencies** and regime-specific **Decision Transformers**, improving **stock performance by 18%** over vanilla sequence-modelling architectures
- Achieved **1.21 Sharpe ratio**, **13% outperformance** vs standard Decision Transformer, and **0.575% monthly alpha** via adaptive policy routing with learned credit assignment across market regimes
- Trained regime-specific Decision Transformers on K-step trajectories with return-to-go conditioning and L-BFGS turnover penalties, using adaptive policy routing based on learned credit assignment [textcolorblueGitHub](#) | [Research Proposal](#)

PIRL – Physics-Informed Reinforcement Learning | *RL, Control Theory, Scientific ML*

- Developed a framework combining physical constraints with deep reinforcement learning for optimal control of nonlinear systems, using physics-guided loss functions to enforce physical invariants during training
- Achieved $8.7\times$ lower training time than Vanilla PINN through smart initialization, eliminating costly λ hyperparameter tuning while improving training efficiency
- Achieved 100% convergence across tested cases versus 33% for Vanilla PINN at $\lambda = 1$, with $R^2 \geq 0.999$ for inlet-velocity prediction

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript, JavaScript, SQL

AI/ML: PyTorch, TensorFlow, JAX, Hugging Face, LangChain, ONNX, Transformers, LLMs, RAG, Fine-tuning, Reinforcement Learning, Decision Transformers, Neural Networks

Backend & Full Stack: React, Node.js, FastAPI, Flask, Express, WebSockets, Prisma

Databases & Infrastructure: PostgreSQL, MongoDB, AWS, Railway, Linux, Git, GitHub

ML Systems & Research: MLOps, Model Serving, Semantic Search, AST Parsing, Bayesian Optimization, AWS SageMaker, Weights & Biases, Lean

CONTESTS & ACHIEVEMENTS

CodeChef: DSA Rating **1770** | Global Rank **1100**

LeetCode: Rating **1600**

Hackathons: **2× College CTE Hackathon Winner** | **Amazon ML Hackathon – All India Rank 53**

Scholarship: BITSians' Day Giving Scholarship

RESEARCH GROUPS & COMMUNITY ENGAGEMENT

- **Member, Society for Artificial Intelligence and Deep Learning (SAIDL)** – Contributing member of the college's student-led AI research organization.
- **Course Mentor, Center for Technical Education (CTE)** – Conducted introductory sessions on machine learning and deep learning foundations, including probability, linear algebra, and optimization.